

phase5 system design

1. Objective of Phase 5

Upgrade system from:

rule-based + signal-based decisions

to:

data-driven, self-improving system

2. What Phase 5 Adds

Core Additions

ML-based credibility (C_{ML}) Memory-based credibility (C_{memory}) ML anomaly detection Feature learning over signals Adaptive weighting

System Now Uses

Rules + Signals + Learning

3. Final Credibility Model

Full Formula

$$C_j(t) = (1 - \alpha) C_{Bayes} + \alpha (C_{ML} + C_{memory})$$

Interpretation

- C_{Bayes} : crowd signal
- C_{ML} : content understanding
- C_{memory} : historical similarity

4. ML Credibility C_{ML}

Definition

$$C_{ML}(j,t) = f(x_j)$$

Feature Vector

$$x_j = [, ,]$$

Concrete Features

TF-IDF / embeddings of text keyword scores early vote ratio interaction velocity

Model Options (Start Simple)

Logistic Regression OR Gradient Boosted Trees

5. Memory-Based Credibility C_{memory}

Formula

$$C_{\text{memory}}(j,t) =$$

Implementation

Step 1: Embed Post

$$v_j = \text{embedding_model}(\text{text}_j)$$

Step 2: Retrieve Similar Posts

$$\text{neighbors} = \text{top_k_similar}(v_j)$$

Step 3: Compute Weighted Average

Similarity

$$\text{Sim}(j,k) = (v_j, v_k)$$

6. ML-Based Anomaly

Formula

$$\text{Anom}_i^{\text{ML}} = (f(x_i))$$

Feature Vector

$$x_i = [S_{\text{post}}, S_{\text{text}}, S_{\text{cred}}, S_{\text{rule}}, S_{\text{user}}, S_{\text{behavior}}]$$

Final Anomaly

$$\text{Anom}_i = (1 - \text{Anom}_{\text{rule}}) + \text{Anom}_{\text{ML}}$$

7. Updated User Weight (Final Form)

$$w_i = T_i (1 - \text{Anom}_i) \text{Exp}_i$$

8. Training Strategy

8.1 Data Collection

store: post text final credibility (label) user behavior

8.2 Labels

For Credibility

$$y_j = \text{final truth (from moderation or consensus)}$$

For Anomaly

label suspicious users manually OR via heuristics

8.3 Training Loop

collect data → train model → deploy → improve

9. Pipeline (Phase 5)

Post created ↓ ML model predicts C_ML ↓ User interactions ↓ Compute: C_Bayes anomaly ↓ Memory lookup
→ C_memory ↓ Combine → C_j ↓ Propagation + Alert

10. What Phase 5 Solves

Problem 1: Early Detection

Before enough votes

Solution:

C_ML gives early signal

Problem 2: Repeated Misinformation

Same false claim appears again

Solution:

C_memory detects pattern

Problem 3: Complex Attacks

Subtle coordinated behavior

Solution:

ML anomaly learns patterns

11. Backend Additions

New Tables

Embeddings

post_embeddings (post_id TEXT, vector VECTOR)

Model Predictions

ml_scores (post_id TEXT, c_ml FLOAT, anomaly_ml FLOAT)

12. Infrastructure

Needed Components

ML inference service Vector database (FAISS / Pinecone) Training pipeline

13. Hyperparameters (Phase 5)

alpha → ML weight gamma → memory weight beta → ML anomaly weight K → neighbors in memory

14. What to Test

Test 1: Early Post

No votes yet

Expect:

C_ML drives credibility

Test 2: Repeated Fake News

similar to past false posts

Expect:

C_memory lowers credibility

Test 3: Subtle Bot Behavior

not obvious from rules

Expect:

ML anomaly detects it

15. Phase 5 Summary

Phase 4: "where matters" Phase 5: "learning matters"

16. Full System Completion

Layer Capability MVP basic Phase 2 user-aware Phase 3 network-aware Phase 4 geo-aware Phase 5 learning system

17. Final Insight

After Phase 5, your system becomes:

self-improving, adaptive, and robust to unknown attacks

18. Reality Check (Important)

Do NOT jump to Phase 5 immediately.

Correct order:

Phase 1 → Phase 2 → Phase 3 → Phase 4 → Phase 5